Fibonacci-Type Sequences

AQA · KS3 Year 7–8

Name: ___________________ Class: _________ Date: _________

Recognise and continue Fibonacci-type sequences, where each term is the sum of the two terms before it. Work forwards and backwards, and find missing terms.

VISUAL EXAMPLE

$$3 \xrightarrow{+4} 7 \xrightarrow{+4} 11 \xrightarrow{+4} 15$$

nth term = 4n - 1

Arithmetic sequence: a constant difference is added each step.

COMMON MISTAKES

- Multiplying the two previous terms instead of adding them.
- Adding the wrong pair of terms — always use the two immediately before.
- For a missing middle term, forgetting that term = (term before) + (term two before).
- Arithmetic slips when the numbers get larger.

METHOD

1. The rule is: each term = the sum of the two terms before it.
2. To continue, add the last two terms together.
3. To find a missing term, use the rule and solve (subtract if needed).
4. To work backwards, subtract: term before = this term - the one before that.
5. Check the rule works across the whole sequence.

WORKED EXAMPLE

Continue: 1, 1, 2, 3, 5, 8, ...
Step 1 — Rule: add the two previous terms.
Step 2 — Next term: 5 + 8 = 13.
Step 3 — Then: 8 + 13 = 21.
Step 4 — (Check backwards: 13 - 8 = 5)
Step 5 — Sequence continues 1, 1, 2, 3, 5, 8, 13, 21, ...

1 Write the next two terms of each sequence. (2 marks)

a) 1, 1, 2, 3, 5, 8, ...

b) 2, 2, 4, 6, 10, ...

HOW TO START

Add the two previous terms to get the next.

For 1, 1, 2, 3, 5, 8: the next term is 5 + 8 = 13.

ANSWERS

(a) = ______

(b) = ______

METHOD 1 Each term = sum of the two before it | 2 Continue by adding the last two | 3 Missing term: use term = prev + prev-prev | 4 Work backwards by subtracting | 5 Check the rule everywhere

2 Each term is the sum of the two terms before it. Write the next term. (2 marks)

a) 3, 5, 8, ...
b) 1, 4, 5, 9, ...

HOW TO START

Each term is the sum of the two before it.
For 3, 5, 8, ...: the next term is 5 + 8 = 13.

ANSWERS

(a) = ____________________

(b) = ____________________

METHOD 1 Each term = sum of the two before it | 2 Continue by adding the last two | 3 Missing term: use term = prev + prev-prev | 4 Work backwards by subtracting | 5 Check the rule everywhere

3 Find the missing term in each Fibonacci-type sequence. (3 marks)

a) 1, __, 3, 5, ...
b) 4, __, 11, 18, ...
c) __, 5, 8, 13, ...

ANSWERS

(a) = ______

(b) = ______

(c) = ______

METHOD 1 Each term = sum of the two before it | 2 Continue by adding the last two | 3 Missing term: use term = prev + prev-prev | 4 Work backwards by subtracting | 5 Check the rule everywhere

4 Using the first two terms given, write the first five terms. (3 marks)

a) Start 1, 3

b) Start 2, 5

c) Start 4, 4

ANSWERS

(a) = ____________________

(b) = ____________________

(c) = ____________________

METHOD 1 Each term = sum of the two before it | 2 Continue by adding the last two | 3 Missing term: use term = prev + prev-prev | 4 Work backwards by subtracting | 5 Check the rule everywhere

5 a) The first two terms are 1 and 2. Write the first six terms. (1 mark) (3 marks)

b) A Fibonacci-type sequence has 3rd term 10 and 4th term 16. Work backwards to find the 1st and 2nd terms. (2 marks)

ANSWERS

(a) = ______

(b) = ______

METHOD 1 Each term = sum of the two before it | 2 Continue by adding the last two | 3 Missing term: use term = prev + prev-prev | 4 Work backwards by subtracting | 5 Check the rule everywhere

6 Each month, the number of rabbit pairs equals the sum of the pairs in the two previous months. Months 1 and 2 each have 1 pair. (3 marks)

a) Find the number of pairs in months 3, 4, 5 and 6. (2 marks)
b) In which month are there first more than 10 pairs? (1 mark)

ANSWERS

(a) = ____________________

(b) = ____________________

METHOD 1 Each term = sum of the two before it | 2 Continue by adding the last two | 3 Missing term: use term = prev + prev-prev | 4 Work backwards by subtracting | 5 Check the rule everywhere

7 A Fibonacci-type sequence has 3rd term 7 and 4th term 11. (4 marks)

a) Find the 1st and 2nd terms. (2 marks)
b) Write the first six terms. (1 mark)
c) Find the 7th term. (1 mark)

ANSWERS

(a) = ____________________

(b) = ____________________

(c) = ____________________

METHOD 1 Each term = sum of the two before it | 2 Continue by adding the last two | 3 Missing term: use term = prev + prev-prev | 4 Work backwards by subtracting | 5 Check the rule everywhere

8 In a Fibonacci-type sequence the 5th term is 22 and the 6th term is 35. (5 marks)

a) Work backwards to find the first four terms. (3 marks)

b) Write the whole sequence (terms 1 to 6). (1 mark)

c) Explain why the terms grow more like a geometric sequence than an arithmetic one. (1 mark)

ANSWERS

(a) = ____________________

(b) = ____________________

(c) = ____________________

METHOD 1 Each term = sum of the two before it | 2 Continue by adding the last two | 3 Missing term: use term = prev + prev-prev | 4 Work backwards by subtracting | 5 Check the rule everywhere

Self-reflection

How confident do you feel with this topic now? (tick one)

Not yet | Getting there | Confident | Really confident

Which question did you find the hardest, and why?

How do I find a missing term in the middle of the sequence?

One thing I will practise before the next lesson:

Teacher key

Q1: a) 13, 21 b) 16, 26

Q2: a) 13 b) 14

Q3: a) 2 (since 1 + 2 = 3, and 2 + 3 = 5) b) 7 (4 + 7 = 11, 7 + 11 = 18) c) 3 (3 + 5 = 8, 5 + 8 = 13)

Q4: a) 1, 3, 4, 7, 11 b) 2, 5, 7, 12, 19 c) 4, 4, 8, 12, 20

Q5: a) 1, 2, 3, 5, 8, 13 b) 2nd term = 16 - 10 = 6; 1st term = 10 - 6 = 4 (sequence 4, 6, 10, 16)

Q6: a) month 3 = 2, month 4 = 3, month 5 = 5, month 6 = 8 b) month 7 (13 pairs)

Q7: a) 2nd term = 11 - 7 = 4; 1st term = 7 - 4 = 3 b) 3, 4, 7, 11, 18, 29 c) 18 + 29 = 47

Q8: a) 4th term = 35 - 22 = 13; 3rd = 22 - 13 = 9; 2nd = 13 - 9 = 4; 1st = 9 - 4 = 5 b) 5, 4, 9, 13, 22, 35 c) the ratio between consecutive terms settles towards a constant value (about 1.618, the golden ratio), so it multiplies by roughly the same factor each time like a geometric sequence, rather than adding a constant like an arithmetic one